

Problem

The spark coil of an automobile ignition system has a 20-mH inductance and a 5-Ω resistance. With a supply voltage of 12 V, calculate: the time needed for the coil to fully charge, the energy stored in the coil, and the voltage developed at the spark gap if the switch opens in 2 μs.

Step-by-step solution

Step 1 of 3

Time taken for the inductor to fully charge is almost equal to five times the time constant of the circuit

$$t_{\text{charge}} = 5 \left(\frac{L}{R} \right)$$

$$t_{\text{charge}} = 5 \times \left(\frac{20 \times 10^{-3}}{5} \right)$$

$$t_{\text{charge}} = 20 \text{ m sec}$$

Step 2 of 3

Energy stored in the coil is

$$W = \frac{1}{2} LI^2$$

$$W = \frac{1}{2} \times 20 \times 10^{-3} \times \left(\frac{V}{R} \right)^2$$

$$W = \frac{1}{2} \times 20 \times 10^{-3} \times \left(\frac{12}{5} \right)^2$$

$$W = \frac{1}{2} \times 20 \times 10^{-3} \times 5.76$$

$$W = 57.6 \text{ mJ}$$

Step 3 of 3

The voltage developed at the spark gap is

$$V = L \frac{\Delta I}{dt}$$

$$V = L \frac{\Delta \left(\frac{V}{R} \right)}{dt}$$

$$V = 20 \times 10^{-3} \times \frac{\left(\frac{12}{5} \right)}{2 \times 10^{-6}}$$

$$V = 20 \times 10^{-3} \times 1.2 \times 10^6$$

$$V = 24 \text{ kV}$$